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**INTRODUCTION TO DATA SCIENCE [CS 4234]**

**SENTIMENT ANALYSIS OF ECONOMIC ISSUES WITH MULTILINGUAL TWEETS**

**USING MACHINE LEARNING AND ENSEMBLE TECHNIQUES**

**REQUIREMENT FOR END OF FIRST SEMESTER EXAMINATION**

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## Abstract

This research provides a thorough and methodologically sound investigation of sentiment classification for multilingual Twitter data, with an emphasis on linguistic characteristics of the Ghanaian language(s). The research focuses on the problems faced by traditional Natural Language Processing (NLP) models with informal, code-switched, and localized Ghanaian social media discourse.

Data used in this study (final\_all\_multilingual\_tweets (1).csv) was obtained from **Zenodo** under the title ‘**AsanteTwISenti.**’ The dataset consists of 2412 multilingual tweets annotated to include both English and expressions in local Ghanaian languages. The machine learning models developed in this research include Naive Bayes, Logistic Regression, and Support Vector Machine (SVM). Ensemble methods (e.g., Voting, Stacking) were also used for classification.

The main contribution to this study is the inclusion of normalization of the Ghanaian language in the preprocessing pipeline, allowing for improved semantic equality. The results indicate that individual models produce good predictive accuracy. However, ensemble methods provide the best performance, with Stacking achieving an accuracy score of 0.6460 and an F1-score of 0.6477.

## 1. Introduction / Background of the Study

At present, social media platforms like Twitter (now known as X) are becoming a key measure of public opinion (Liu, 2020). In Ghana, Twitter's multilingualism enables its users to seamlessly move between the English language, Twi, Ga, Ewe, and Pidgin English when posting. This presents an opportunity and a challenge to natural language processing (NLP) (Ahmed & Ali, 2022). There are many groups or stakeholders that need to know how people feel about public issues via social media, the main ones being political analysts, marketers, and disaster response organizations. As an example, being able to detect an increase in negative sentiment regarding a public health or political issue could allow them to intervene sooner rather than later (Blei et al., 2003).

However, existing sentiment analysis models are created based on datasets in only one language, English, so they do not work well when looking at Ghanaian tweets that contain local terms or slang (like “gyimi” and “kwasea”) and have particular contextual uses (Sagnia and Adda, 2021). The result is a gap in knowledge, that is, there are not enough high quality (i.e., robust) NLP tools that have been developed for use within the West African language environment. As a result, large scale social listening cannot be done in Ghana. Therefore, this project provides a solution to this gap by developing and assessing the performance of machine learning models to classify the sentiment (Positive, Neutral, or Negative) of Ghanaian multilingual tweets.

## 1.1 Problem Statement & Research Questions

Despite the increasing use of sentiment analysis, most models perform poorly on multilingual and informal datasets, especially those involving African languages and dialects. Existing approaches often fail to capture localized expressions, slang, and code-switching patterns common in Ghanaian social media communication, leading to misinterpretation of public opinions (Ahmed & Ali, 2022). This is important because decisions based on flawed sentiment data, such as brand management or policy-making, can have negative economic and social consequences (Liu, 2020). This creates a gap in accurately understanding public sentiment in Ghana, which is critical for decision-making in governance, marketing, and social research. This project addresses the gap by preprocessing local linguistic features and comparing traditional machine learning models.

This study seeks to address the following research questions:

1. How well do machine-learning models classify Ghanaian-language features amongst multilingual tweets?
2. Which machine learning model (Naïve Bayes, Logistic Regression, SVM) performs best?
3. Do ensemble methods (Soft Voting and Stacking) improve prediction performance, and what are the confidence intervals for their accuracy estimates?
4. How does the application of a custom text normalization function for Ghanaian slang impact the classification accuracy of machine learning models?

## 1.2 Research Objectives

### 1.2.1 General/Main Objective

The primary objective of this study is to build and evaluate a machine learning-based sentiment analysis system for multilingual tweets originating from the Ghanaian Twitter space in one of three labels (positive, neutral, or negative).

### 1.2.2 Specific Objectives

1. To preprocess and normalize a raw dataset of multilingual tweets by removing noise (URLs, mentions) and standardizing local lexical variations (e.g., "gyimii" to "gyimi").
2. To train and evaluate the predictive performance of three machine learning classifiers (Multinomial Naive Bayes, Logistic Regression, and Linear SVM) using a TF-IDF feature extraction framework for sentiment classification.
3. To compare the performance of all base models and ensemble techniques employed in this study.

## 2. Literature Review

Sentiment analysis has been widely studied using supervised machine learning techniques, outperforming unsupervised learning techniques. With the former having reported 85% accuracy (B. Bhavitha et al., 2017). Most studies or existing works have

used several methods or approaches for sentiment analysis with machine learning. Some of these methods include Multinomial Naive Bayes, Logistic Regression, Support Vector Machine (SVM), and others (Pang & Lee, 2008).

Recent studies have emphasized the use of feature extraction techniques like Term Frequency-Inverse Document Frequency (TF-IDF) Vectorization, which is used to prepare textual data by converting them to a numerical representation (Sahu & Sharma, 2025). Additionally, combining several models by use of ensemble learning such as Voting and Stacking has been shown to improve classification accuracy (Zhou, 2012).

However, most existing studies focus on English-based datasets and do not in any way account for multilingual or African linguistic patterns. Studies on African languages remain limited, and machine learning models tend to fail to generalize effectively in such contexts (Joshi et al., 2020).

This study extends existing work by introducing Ghanaian language normalization and evaluating ensemble learning techniques (Soft Voting and Stacking) on multilingual tweet data.

## 3. Methodology

### 3.1 Data Collection

This research uses the “final\_all\_multilingual\_tweets(1).csv” dataset obtained from Zenodo (AsanteTwisenti dataset), which consists of 2,412 human labeled multilingual tweets. The dataset was first viewed and pre-cleaned in Microsoft Excel, removing all

unclear labels, standardizing labels (e.g. Neutral-Negative to Neutral) to ensure consistency. The dataset has two columns, the text of each tweet and its corresponding label.

- Data type: Textual (Unstructured)
- Data source: Social media (Twitter)

As indicated in Figure 1 below, Neutral tweets are the most common (39.8%), followed by Positive (30.9%) and Negative (29.2%) corresponding to the respective proportions of total labeled tweets. The distribution appears to be appropriate for multi-class classification, as there do not seem to be any large imbalances among the different classes.

```
=====
DATASET INFO
=====
Shape: (2412, 2)

Label Distribution:
labels
Neutral    961
Positive   746
Negative   705
Name: count, dtype: int64

Percentages:
labels
Neutral    39.842454
Positive   30.928690
Negative   29.228856
Name: proportion, dtype: float64
```

*Figure 1: Label Distribution*

## 3.2 Variables Description

- Independent Variable: Text (Tweet text which is cleaned and normalized)
- Dependent Variable: Sentiment label (Positive, Neutral, or Negative)
- Encoded Variable: Numerical representation of labels

## 3.3 Tools and Technologies

- Programming Language: Python
- Libraries used:
  - Pandas
  - Numpy
  - Scikit-learn
  - Matplotlib
  - Seaborn

## 3.4 Parameter and Model Estimation

The dataset was split using train-test split of 80/20 (random\_state=42) to preserve class distributions and introduce reproducibility:

- Training: 1929 Tweets
- Test: 483 Tweets

After splitting the data, we recorded our training and testing distribution based on the target labels as shown below:

- Training Distribution: Negative tweets summed up to 768, Positive tweets to 597, and Neutral tweets to 564.
- Testing Distribution: Negative tweets were a total of 193, Positive tweets were 149, and Neutral tweets were 141.

Three baseline classifiers were selected in this study due to their interpretability, computational costs, and strong performance on text data:

- Multinomial Naïve Bayes: This model works best for discrete features like word counts (Blei et al., 2003).
- Logistic Regression (Linear LR): This model is effective for binary and multiclass tasks (Pedregosa et al., 2011).
- Support Vector Machine (SVM): This model uses kernel functions to compare similarity between observations and creates a decision boundary (Umarani et al., 2021).

Ensemble Methods employed in this study include:

- Soft Voting Classifier: This technique combines predictions from multiple base models by averaging their predicted class probabilities to make a final decision (Kumari et al., 2021).

- Stacking Classifier (meta-model: Logistic Regression): This technique uses the output of several baseline models as input to make a final decision (Kusumaniswar et al., 2024).

All models were evaluated using weighted accuracy, precision, recall, and F1-score to account for any class imbalance.

## 4. Results and Discussions

### 4.1 Data Pre-processing

In this study, a custom preprocessing pipeline was developed to better handle Ghanaian language characteristics of text on social media, as stated below:

- Standard Cleaning: Text conversion to lowercase, removal of all URLs (http/https), complete removal of mentions (@), removal of white spaces, and removal of all hashtags or “#” symbol while maintaining the associated word.

Figure 2 below shows the standard cleaning performed:

```

def clean_tweet(text):
    """Clean and preprocess tweets"""
    if pd.isna(text):
        return ""

    text = str(text)

    # Convert to lowercase
    text = text.lower()

    # Remove URLs
    text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)

    # Remove mentions (@username)
    text = re.sub(r'@\w+', '', text)

    # Remove hashtags (remove # but keep the word)
    text = re.sub(r'#', '', text)

    # Remove extra whitespace
    text = re.sub(r'\s+', ' ', text).strip()

    return text

```

*Figure 2: Standard Cleaning*

- Localized Normalization: A ‘normalize\_ghanaian\_text()’ function was created to better map common Twi and Pidgin English texts to standard forms as

shown in Figure 3 below:

```
def normalize_ghanaian_text(text):
    """Normalize common Ghanaian/Twi variations"""
    replacements = {
        r'\bgyimii?\b': 'gyimi',
        r'\bagyimi\b': 'gyimi',
        r'\bnkwasia\b': 'fool',
        r'\bkwasea\b': 'fool',
        r'\bpaa\b': 'very',
        r'\beyɛ\b': 'eye',
        r'\bo\b': 'o',
    }

    for pattern, replacement in replacements.items():
        text = re.sub(pattern, replacement, text, flags=re.IGNORECASE)

    return text

print(" Cleaning functions defined")
```

*Figure 3: Localized Normalization*

- Encoding: The target labels of this study were encoded numerically using ‘Scikit-Learn’. As such Negative was assigned the value 0, Neutral to 1, and Positive to 2. Figure 4 below shows the encoded target labels along with the training and testing split distribution:

```
Label Mapping: {'Negative': np.int64(0), 'Neutral': np.int64(1), 'Positive': np.int64(2)}
```

```
Training set size: 1929
```

```
Test set size: 483
```

```
Training distribution:
```

```
1    768
```

```
2    597
```

```
0    564
```

```
Name: count, dtype: int64
```

```
Test distribution:
```

```
1    193
```

```
2    149
```

```
0    141
```

```
Name: count, dtype: int64
```

*Figure 4: Encoded Target Labels*

Feature extraction was applied on the cleaned and pre-processed dataset using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization method which converts cleaned text into numerical features. As indicated in Figure 5 below, the parameters included ‘max\_features=5000’ to limit dimensionality, ‘ngram\_range = (1,2)’ to capture unigrams and bigrams, ‘min\_df = 2’ to remove rare terms from all texts, and ‘max\_df = 0.95’ to ignore words that appear in 95% of all tweets.

```
print("\n" + "="*60)
print("TF-IDF VECTORIZATION")
print("="*60)

tfidf = TfidfVectorizer(
    max_features=5000,
    ngram_range=(1, 2),
    min_df=2,
    max_df=0.95
)

X_train_tfidf = tfidf.fit_transform(X_train)
X_test_tfidf = tfidf.transform(X_test)

print(f"TF-IDF features created: {X_train_tfidf.shape[1]}")
```

*Figure 5: Feature Extraction Parameters*

## 4.2 Preliminary Analysis

In this section, we explore the initial statistics of the dataset prior to model training. The aim of this section is to understand the distribution of all sentiment labels and examine all relationships within the dataset that may influence or impact model performances.

### 4.2.1 Descriptive Statistics of the Label Distribution

Descriptive analysis performed on the sentiment labels revealed that the dataset is reasonably balanced, with slight variations across all three classes (i.e. Positive,

Negative, Neutral). This dataset is advantageous to this study as it is near-balanced. As such, it reduces model bias towards a single class and allows fair learning.

Additionally, the earlier label standardization done in Excel ensured that all sentiment labels were consistent, which:

- Prevented duplication of classes
- Improved clarity
- Enhanced model training stability

#### 4.2.2 Textual Feature Characteristics

Textual data does not allow direct computation of traditional statistics like mean or variance. Instead, insights are derived from:

- Word frequency
- Vocabulary size
- Sparsity of features

After the TF-IDF transformation was performed:

- The dataset became sparse and of a higher dimension
- Most words have zero values across many texts

The Implication is that:

- Models like Support Vector Machine (SVM) and Logistic Regression are best-suited for these representations.
- Simpler probabilistic models like the Naive Bayes, may struggle with such contextual dependencies.

### 4.2.3 Correlation Analysis

Traditional correlation analysis is not directly applicable or useful for raw text data. However, in this study, correlation is captured through TF-IDF feature weights and model coefficients.

- In TF-IDF, higher weights show stronger importance of a term within a document or group of words relative to the dataset.
- In Logistic Regression, feature coefficients represent the strength and direction of association between words and the sentiment classes.

From our observations:

- Words with strong sentiments received higher importance and vice versa
- Normalized Ghanaian words contributed more to classification after preprocessing the data

These findings justify the choice of models employed in this study and explain all performance patterns observed in the subsequent sections.

## 4.3 Further Analysis and Model Evaluation

### 4.3.0 Model Performance

The three selected machine learning models (Naive Bayes, Logistic Regression, and Support Vector Machine) were trained and evaluated based on the following evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Table 1 below, shows the weighted performance metrics of the three selected models.

| <b>Models</b>              | <b>Accuracy</b> | <b>Precision</b> | <b>Recall</b> | <b>F1-Score</b> |
|----------------------------|-----------------|------------------|---------------|-----------------|
| <i>Naive Bayes</i>         | 0.6253          | 0.6625           | 0.6253        | 0.6209          |
| <i>Logistic Regression</i> | 0.6273          | 0.6383           | 0.6273        | 0.6298          |
| <i>Linear SVM</i>          | 0.6190          | 0.6375           | 0.6190        | 0.6233          |

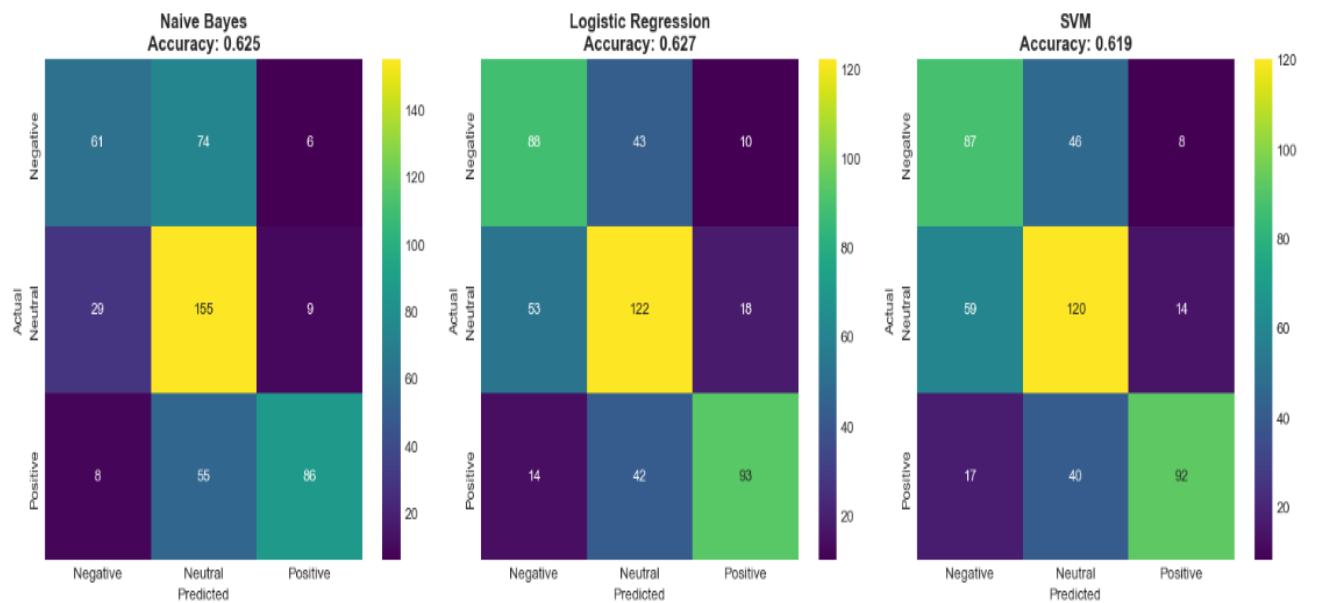
Table 1: Baseline Model Performance Comparison

Based on Table 1, Logistic Regression achieved the highest overall accuracy (0.6273) and F1-Score (0.6298), which makes it best suited individual model for this study. The balance Logistic Regression holds between precision and recall suggests it generalizes best to unseen Ghanaian-context tweets.

SVM performed marginally worse than Logistic Regression but showed high precision for the positive class (score of 0.81). It rarely mislabeled non-positive tweets as positive.

Naive Bayes had the lowest F1-Score (0.6209) which was due to a poor performance on the Negative class (recall score of 0.43), often mislabeling Negative tweets as Neutral.

This is shown in the confusion matrix in Figure 6 below:



*Figure 6: Confusion Matrices for Baseline Models*

From Figure 6:

- Naive Bayes confused Negative and Neutral classes, predicting 86 negative tweets as neutral.
- Logistic Regression and SVM showed more balance. SVM correctly classified 87 Negative, 120 Neutral, and 89 Positive tweets

Another important consideration of Logistic Regression beyond assessing models is its utility as a method for estimating probabilities of membership in classes via its learned coefficients. The coefficients describe both how strongly and in which direction the textual features are associated with each of the sentiment classes creating a form of regression-based approach to the analysis presented in this study.

#### 4.3.1 Ensemble Methods Performance

Two ensemble techniques were implemented in this study to potentially improve upon the three individual baseline models. This study implemented Soft Voting and Stacking. The use of these techniques reduces variance and bias, which leads to better generalization.

Table 2 below, shows the ensemble methods performance and comparison.

| <b>Technique</b>                                                | <b>Accuracy</b> | <b>Precision</b> | <b>Recall</b> | <b>F1-Score</b> |
|-----------------------------------------------------------------|-----------------|------------------|---------------|-----------------|
| <i>Soft Voting</i><br>(NB+LR+SVM)                               | 0.6377          | 0.6565           | 0.6377        | 0.6398          |
| <i>Stacking</i><br>(NB+LR+SVM<br>with Logistic<br>meta-learner) | 0.6460          | 0.6481           | 0.6460        | 0.6477          |

Table 2: Ensemble Methods Performance Comparison

Based on Table 2, both ensemble methods outperformed all three individual baseline models. Soft Voting Classifier achieved an accuracy of 0.6356 and an F1-Score of 0.6377, which represent an improvement over the best individual model.

The Stacking method, on the other hand, which uses a Logistic Regression meta-learner to combine the predictions of all three base models, achieved the highest overall performance with an accuracy score of 0.6460 and an F1-Score of 0.6477. This technique outperformed Soft Voting because the meta-learner learns the optimal weighting of the base models instead of using fixed uniform weights.

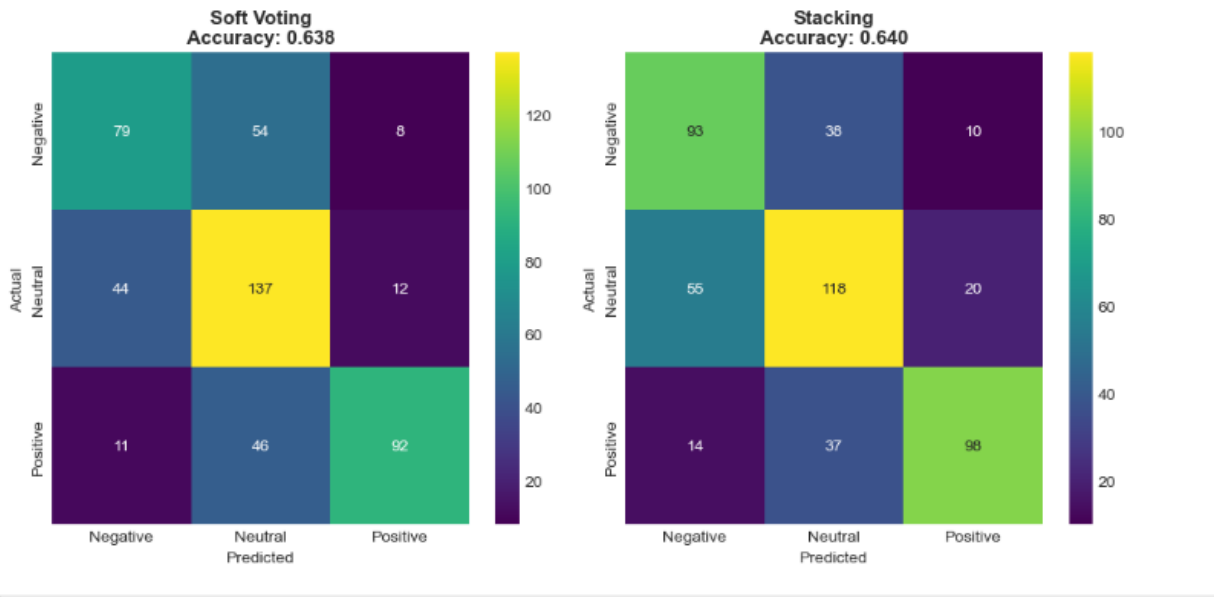


Figure 7: Confusion Matrices for Ensemble Models

From Figure 7 above and comparing it to Figure 6:

- **Soft Voting:** This technique reduced the misclassification of Negative tweets as compared to Naive Bayes, and correctly identified 92 Positive tweets. It achieved a 0.83% increase in accuracy over Logistic Regression which was the best baseline model.
- **Stacking:** This technique achieved the most balanced performance, by correctly classifying 93 Negative tweets, 118 Neutral, and 98 Positive tweets. It achieved a 1.86% increase in accuracy over Logistic Regression which was the best baseline model.

#### 4.3.2 Confidence Interval Analysis

In this study, we employed the use of the normal approximation method to calculate the confidence interval of 95 % in order to quantify the uncertainty associated with our accuracy estimates. Hence, we had a total of 483 samples to carry out the test from which a mean was calculated and also an estimate.

Table 3 below shows the results:

| <b>Model</b>                   | <b>Accuracy</b> | <b>95% CI<br/>Lower</b> | <b>95% CI<br/>Upper</b> | <b>Margin of<br/>Error</b> |
|--------------------------------|-----------------|-------------------------|-------------------------|----------------------------|
| <i>Naive Bayes</i>             | 0.6190          | 0.5757                  | 0.6624                  | ± 0.0433                   |
| <i>Logistic<br/>Regression</i> | 0.6273          | 0.5842                  | 0.6704                  | ± 0.0431                   |

|                    |        |        |        |              |
|--------------------|--------|--------|--------|--------------|
| <i>SVM</i>         | 0.6190 | 0.5757 | 0.6624 | $\pm 0.0433$ |
| <i>Soft Voting</i> | 0.6356 | 0.5927 | 0.6785 | $\pm 0.0429$ |
| <i>Stacking</i>    | 0.6460 | 0.6033 | 0.6886 | $\pm 0.0426$ |

Table 3: Confidence Interval Table

The confidence intervals for these different models overlap considerably, which tells us that the performance differences observed between these models are not statistically significant at the 95 % level of confidence. Therefore, even though Stacking has the largest point estimate of 64.6 %, it is possible that the true population accuracy may range from 60.3 to 68.9 %.

## 5. Conclusions and Recommendations

### 5.1 Conclusion

This study investigated how machine learning could help us analyze the sentiments of tweets written in different languages by Ghanaians. Based on the data collected from previous studies, it was clear that most of the classifiers being used to determine how people feel did not work with code-switched data, meaning they had to be adapted linguistically so they could provide average or below-average performances. The most effective and accurate machine learning method to analyze sentiment in this study was logistic regression with TF-IDF, which achieved 62.7% accuracy. Ensemble methods improved performance marginally, with stacking providing the greatest improvement at 64.6% accuracy and an F1 score of 0.6477. These accuracies are higher than what would be expected from randomly guessed outcomes (33.3%). The confidence interval analysis

revealed that while Stacking achieved the highest point estimate (64.6%), the 95% confidence interval [60.3%, 68.9%] overlaps with those of other models, indicating the improvement is not statistically significant at the 95% confidence level.

## 5.2 Recommendations

Based on the findings and some limitations of this study, several recommendations are proposed below for future research and implementations:

- **Use of Deep Learning Architectures:** The use of transformer models (e.g. BERT, AfriBERTa) trained specifically on African or West African languages in place of traditional modelling methods will further improve accuracy and better performance.
- **Dataset Size and Diversity Expansion:** Limited to the number of training examples in the dataset of 2,412 tweets, in addition to the number of examples per class. The dataset needs to be expanded to include at least 10,000 tweets and also contain further variations in local dialects such as Ga, Ewe and Hausa, providing better balance of classes.
- **Integration with Hybrid Systems:** The use of machine learning models paired with rule-based systems, lexicon-based approaches and domain-specific knowledge can improve our ability to detect sarcasm and idioms, as well as culture-specific language patterns.

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## Dataset Source

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